THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Particulate Nature of Matter** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** tiny particles	Matter consists of very small particles.
2	**(c)** melting	Fusion.
3	**(d)** increase in temperature	Warmer liquids evaporate faster.
4	**(d)** sugar dissolves in water without raising the water level much	Sugar particles occupy spaces between water particles.
5	**(c)** evaporation	Surface process.
6	**(d)** spread out in a windy place	More surface and wind.
7	**(b)** constantly moving	Particles are always in motion.
8	**(c)** melting point	Melting point.
9	**(c)** cooling	Fastest particles leave, so cooling.
10	**(c)** diffusion	Particles of ink mix with water particles.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Matter is made of tiny particles (1); Particles move and have spaces between them (1).	1+1
12	Evaporation occurs at any temperature; boiling at a fixed temperature (1); Evaporation is at the surface; boiling throughout the liquid (1).	1+1
13	Condensation (1); Water vapour in air cools and becomes liquid (1).	1+1
14	Spontaneous mixing of particles of two substances (1); Smell of perfume spreading in a room (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Dissolve a spoon of salt/sugar in a beaker of water (1); Level of water hardly rises (1); Solute particles fit in the gaps between water particles (1).	1+1+1
16	Fastest particles escape from the liquid (1); Remaining particles have lower energy (1); Temperature of the liquid falls (1).	1+1+1
17	Water vapour in air touches the cold glass (1); It loses energy and condenses (1); Droplets form (1).	1+1+1
	OR: Put a crystal of potassium permanganate in water (1); Colour spreads slowly (1); Particles move and mix on their own (1).	1+1+1
18	Cool it (1); Compress it (1); Particles come closer and attract each other (1).	1+1+1
	OR: Temperature rises when heating (1); Flat at 0: melting (1); Flat at 100: boiling (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Dissolving sugar (1); Spreading of ink (1); Smell of perfume (1); Spaces between particles (1); Particles in constant motion (1).	1+1+1+1+1
	OR: Wrap cotton on the bulb (1); Dip in water, then in acetone (1); Note the temperature fall (1); Acetone gives more cooling (1); Faster evaporation leads to more cooling (1).	1+1+1+1+1
20	Shape (1); Volume (1); Compressibility (1); Particle arrangement (1); Forces between particles (1).	1+1+1+1+1
	OR: Particles move randomly (1); They collide with container walls (1); Collisions exert force (1); More collisions give more pressure (1); Heating increases the pressure (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Particulate Nature of Matter** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 0 degrees Celsius	Ice melts at 0.
2	**(c)** cools the body	Cooling by evaporation.
3	**(d)** gases	Gas particles move quickly with big gaps.
4	**(b)** 100 degrees Celsius	Water boils at 100.
5	**(b)** amount of water vapour in the air	Moisture content.
6	**(c)** solids	Strong forces hold them in place.
7	**(c)** condensation	Gas to liquid.
8	**(d)** decreases	Air is already saturated.
9	**(d)** liquid	Liquids take the shape of the container.
10	**(d)** sublimation	Camphor and naphthalene.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It evaporates quickly (1); Takes heat from the skin (1).	1+1
12	Particles move faster at higher temperature (1); Diffusion is faster (1).	1+1
13	Particles move rapidly in all directions (1); Forces between them are very weak (1).	1+1
14	Direct change of a solid into a gas (1); Camphor, naphthalene, dry ice (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Solids: close, fixed, vibrate (1); Liquids: close, can slide past (1); Gases: far apart, move freely (1).	1+1+1
16	Melting ice absorbs a large amount of heat from the drink (1); More heat is taken from the drink (1); Water at 0 degrees Celsius absorbs heat only by warming up (1).	1+1+1
17	Camphor, naphthalene, dry ice (1); Dew, mist, fog (1); Water droplets on glass (1).	1+1+1
	OR: Solid: fixed shape (1); Liquid: fixed volume, no fixed shape (1); Gas: no fixed shape or volume (1).	1+1+1
18	Steam carries extra heat that was absorbed during vaporisation (1); It gives this heat out on condensing on the skin (1); More heat is transferred to the skin (1).	1+1+1
	OR: Particles move faster when hot (1); Faster diffusion (1); Hot vapour spreads quickly (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Melting (1); Freezing (1); Evaporation/boiling (1); Condensation (1); Sublimation (1).	1+1+1+1+1
	OR: Strong forces in solids (1); Weaker in liquids (1); Very weak in gases (1); Heating weakens the forces (1); Cooling strengthens them (1).	1+1+1+1+1
20	Surface process at any temperature (1); Temperature, surface area, wind, humidity (1); Drying clothes (1); Cooling of earthen pot (1); Sweating (1).	1+1+1+1+1
	OR: Same cloth and water (1); Vary one factor only (1); Measure drying time (1); Higher temperature and wind dry faster (1); Humidity slows it (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Particulate Nature of Matter** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** water evaporates through tiny pores of the pot	Evaporation cools.
2	**(c)** gas	Gas fills the whole container.
3	**(c)** camphor	Camphor changes directly.
4	**(d)** increasing pressure and lowering temperature	Compress and cool.
5	**(d)** gases	Large gaps between particles.
6	**(c)** remains constant	Energy used to change state.
7	**(b)** liquid under high pressure	Gas liquefied by pressure.
8	**(d)** solids	Particles are tightly packed.
9	**(b)** overcome the forces of attraction between particles	The temperature stays constant during the change of state.
10	**(a)** carbon dioxide	It sublimes.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Large spaces between particles (1); Particles can be pushed closer (1).	1+1
12	Particles are tightly packed (1); Very little space is available (1).	1+1
13	Heat is used to break forces between particles (1); Not to raise the temperature (1).	1+1
14	Gas is compressed under high pressure (1); It turns into a liquid (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Temperature (1); Surface area and wind speed (1); Humidity (1).	1+1+1
16	Smell of cooking (1); Perfume spreading (1); Tea dispersing in water (1).	1+1+1
17	Particles gain energy and move faster (1); At boiling point they break free throughout (1); Bubbles of vapour form (1).	1+1+1
	OR: Low humidity speeds evaporation (1); More cooling is achieved (1); In humid weather it is less effective (1).	1+1+1
18	Melting point: solid to liquid (1); Boiling point: liquid to gas (1); Both are fixed for a pure substance (1).	1+1+1
	OR: Compressibility: solids none, liquids little, gases high (1); Density: highest in solids typically (1); Shape: solids fixed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Heat energy supplied is used to overcome forces between particles (1); It is not used to increase the motion of particles (1); So the temperature stays constant during melting (1); and also during boiling (1); Temperature rises again after the change of state is complete (1).	1+1+1+1+1
	OR: Camphor changes to vapour (1); Salt does not (1); Heat mixture in a china dish with an inverted funnel (1); Camphor collects on funnel (1); Pure salt remains (1).	1+1+1+1+1
20	Cooling lowers particle energy (1); Pressure pushes particles closer (1); LPG is liquefied under pressure (1); Dry ice sublimes at normal pressure (1); Gases can be liquefied (1).	1+1+1+1+1
	OR: Hot vapour near the spout is invisible (1); Cooler air makes it condense (1); Tiny droplets form (1); Droplets scatter light as a white cloud (1); This is condensation (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.